MAXWELL MCINNIS | B.ENG

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PROFESSIONAL SUMMARY

Resourceful engineer with hands on experience in product management, research and development, automation and system optimization. Led teams to deliver data driven solutions from concept to deployment with focus on usability, performance and impact. Experienced in mechanical automation design, software integration, cloud and web development, agile methodologies and continuous improvement. Skilled in stakeholder engagement and cross functional collaboration. Passionate about leveraging emerging technologies to drive innovation and efficiency.

AREAS OF EXPERTISE

- ✓ Data visualization (Excel, MATLAB, Python, Power BI)
- ✓ Embedded systems (ESP32, Arduino) & rapid prototyping
- ✓ Sensor integration and circuit design
- ✓ B2B technical sales & client onboarding
- ✓ Workflow optimization and process design
- ✓ Web design with Python, Django, Gunicorn, Javascript, Nginx, HTML & CSS
- ✓ Experimental design and validation
- ✓ Inventory tracking & hardware lifecycle control
- ✓ Technical project planning execution & sprint execution
- ✓ Mechanical design (SolidWorks CSWA)
- ✓ FEA and simulation (ANSYS tools)
- ✓ Multi-disciplinary mechatronics design
- ✓ Team leadership & resource coordination
- ✓ Programming in C, C++, Python, HTML, JavaScript, & CSS
- ✓ ERP system setup and customization
- ✓ Strategic operations & resource forecasting

CAREER HIGHLIGHTS

Leadership: Held management roles at three companies, applying lean methodologies and standardizing processes to improve efficiency and delivery reliability.
Academics: Published co-author in peer-reviewed research (*Water Environment Research*, 2024) focused on sustainable bioproduct development.
Continuous learning: Graduated with honours and distinction in Biomedical Engineering and will begin a part-time MBA in Fall 2025 to build business, leadership and strategic skills.

PROFESSIONAL EXPERIENCE

Operations Manager, Kraken Sense (BioTech Start-Up), *Oakville, ON* | April 2025 – Current

Oversaw full Kraken production with SOPs to meet targets and quality; led next-gen R&D and supported sales, marketing, and funding via technical presentations and customer engagement.

- Engineered and optimized an automated qPCR pathogen sensing system, **reducing failure rates by over 32×** and enabling mass production scalability; integrated fluorescence optics, mechatronics, Python software, embedded C firmware, hydraulic pumps, custom PCBs, and linear actuators for precise, high-reliability operation.
- Managed Kraken production operations and staffing applying lean methodologies such as 5S, value stream mapping, and Kaizen events to **reduce cycle time by 30%** and improve overall equipment effectiveness by **20%**
- **Generated over $400K in Kraken revenue** through lead generation client outreach and technical sales with marketing.
- Modified open-source ERP (InvenTree) with Python, Django & React for real-time tracking, **driving 347% revenue growth per unit and 14× on-time delivery**, boosting customer satisfaction.
- Engineered multiple automated jigs for quality control of Kraken consumable sensors **reduced QC costs by over 700%** and enabled non-technical staff to run tests reliably.

Web Designer, Self |-Employed, | April 2025 – Current

Independent web designer delivering scalable, full-stack websites for small businesses to enhance online presence and drive growth.

- Delivered 6+ custom websites for small businesses, increasing client visibility and **lead generation by up to 60%**
- Developed full-stack solutions using JavaScript, Python, Django, HTML/CSS, improving site performance and usability
- Deployed to DigitalOcean with Nginx and Gunicorn, achieving **99.9% uptime** and reducing downtime-related inquiries by 80%

Engineering Administrator, Walinga, *Guelph, ON* | January 2025 – April 2025

Engineering Administrator at Walinga overseeing R&D coordination, workflow automation, staff allocation, reporting systems and SR&ED documentation to accelerate project delivery and secure tax credits.

- Streamlined workflows and systems, cutting delays and improving collaboration across engineering teams.
- Managed schedules and allocations for 15+ staff, **accelerating project delivery by 20%**.
- Led interviews with accountants and directly represented Walinga in CRA audits, securing over **$250K in SR&ED tax credits**
- Automated engineering scheduling using MariaDB and Power BI, transforming a **4-hour manual task into a zero-time workflow** & creating live KPI dashboards to track output, resourcing, and progress across all active projects.

Engineering Researcher, University of Guelph, *ON*| April 2024 – April 2025

Conducted hands-on research in biomedical engineering with a focus on wearable sensors, biomechanics, and sustainable materials.

- Designed and tested sensor-based wearable devices for gait correction, using ESP32, IMUs, and vibration feedback systems.
- Developed signal-processing algorithms for real-time feedback and calibrated IMU drift using sensor fusion methods.
- Led experimentation schedules and coordinated tasks across a small research team to ensure timely publication outcomes.
- Contributed to sustainability-focused projects, transforming Miscanthus wastewater into materials like vegan leather and detergents, and presenting findings at research conferences and successfully coauthoring a published paper.

Accountant/Bookkeeper, Concorde Accounting & Tax, *Guelph, ON* | May 2024 - August-2024

Supported client intake, documentation, and administrative workflows during peak tax season. Ensured accuracy, confidentiality, and speed in a high-volume, deadline-driven environment.

- Processed tax documents for **100+ clients**, verifying accuracy and completeness under tight turnaround times.
- Maintained organized digital and paper records while handling sensitive financial data with discretion.
- Coordinated client communications, appointment scheduling, and file tracking using in-house systems.

Municipal Staff, City of Pickering, Pickering, ON | 2022-2023

Progressed from front-line service to operations leadership, driving efficiency and team performance across multiple locations.

- Advanced from entry-level to trusted lead in field operations, supporting infrastructure upkeep and team coordination.
- Maintained city parks and public assets, ensuring safety, quality, and aesthetic standards across 20+ locations.
- Mentored seasonal staff and optimized task delegation to streamline daily site operations and reduce downtime.
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Operations Manager & Crew member, Bartlett Family Restaurants, Pickering, ON | September 2018 – July 2021

Progressed from front-line service to operations leadership, driving efficiency and team performance across multiple locations.

- Delivered fast, accurate service **to 500+ customers daily**, ensuring high satisfaction scores.
- **Trained and developed 20+ staff,** improving team productivity and service consistency
- Optimized scheduling and workflows, cutting labor costs and **reducing service times during peak hours by 10%.**
- Helped optimize intake procedures by identifying bottlenecks and standardizing form processing.

EDUCATION

B. Eng Biomedical Engineering 2021-2025

University of Guelph, *Guelph, ON*